Review of Field Weakening Control Strategies of Permanent Magnet Synchronous Motors

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***Abstract*—Due to high power density, high efficiency, and accurate control performance, permanent magnet synchronous motors (PMSMs) have been widely adopted in equipment manufacturing and energy transformation fields. To expand the speed range under finite DC-bus voltage, extensive research on field weakening (FW) control strategies has been conducted. This paper summarizes the major FW control strategies of PMSMs, which are categorized into calculation-based methods, voltage closed-loop control methods, and model predictive control related methods. The existing strategies are analyzed and compared according to performance, robustness, and execution difficulty, which can facilitate the implementation of FW control.**

***Index Terms*—Calculation-based methods, Field weakening control, Model predictive control, Permanent magnet synchronous motor, Overmodulation, Voltage closed-loop control.**

I. Introduction

OWING to high control performance and high power density, permanent magnet synchronous motors (PMSMs) have broad application prospects in electric drive [1]-[4]. When the motor operates below base speed, maximum torque per ampere (MTPA) control is typically used to reduce motor losses [5]. When above base speed, the motor's back electromotive force (EMF) increases with speed. Once the inverter output voltage saturates, the motor speed cannot further increase. In this case, a field weakening (FW) control strategy is needed to generate FW current that can counteract the motor's back EMF, thereby expanding the motor's speed range. According to different FW control theories, these strategies can be divided into calculation-based methods, voltage closed-loop control methods, and model predictive control (MPC) related methods shown in Fig. 1.

Due to the limited computational capability of early microcontroller units (MCUs), offline calculation, i.e., lookup table (LUT) methods, were prevalent FW control algorithms [6]-[7]. According to the table input variables, they can be divided into torque-speed LUTs [8]-[14], torque-flux LUTs [15]-[17], and multivariate LUTs [18]-[21]. With the advancement of chip manufacturing technology, the enhanced computational capability of MCUs allows for the online calculation of FW current, enabling more versatile FW control strategies through optimization of the computation process and results [22]-[31]. Process optimization aims to reduce the number of algorithmic calculations or iterations using mathematical methods, thereby reducing the computational burden on the MCUs [32]-[45]. Result optimization improves the accuracy of online calculation by considering non-ideal motor factors through correction or compensation methods [46]-[55]. To eliminate the dependency on motor models, voltage closed-loop FW control strategies have gained favor. Based on the number of regulators, these strategies can be classified into dual-regulator [56]-[86] and single-regulator [87]-[96] control structures. In the dual-regulator structure, FW current or current angle can be adjusted by voltage difference or duty cycle difference, enabling smooth switching between FW control and MTPA control. Besides, the single-regulator structure fixes the output voltage amplitude and directly distributes the d-q axis voltages to avoid FW output voltage saturation. To further enhance the dynamic and steady-state performance, the MPC related methods have been widely investigated, which focus on the effects of system nonlinear constraints [97]-[99]. Based on the characteristics of the output voltage vector, they can be

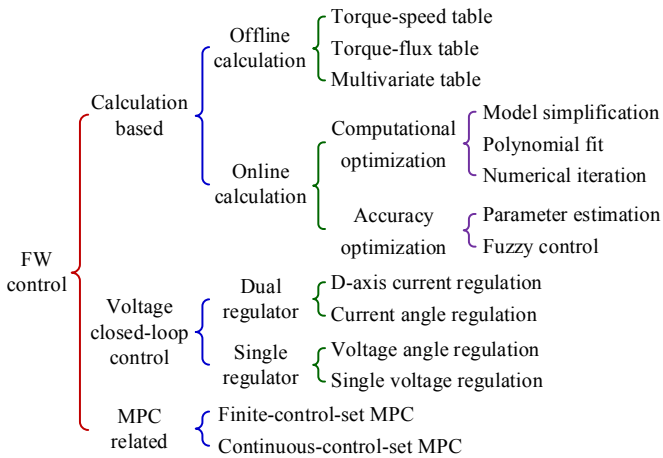


Fig. 1. Classification of FW control strategies.

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divided into finite-control-set (FCS) [100]-[103] and continuous-control-set (CCS) methods [104]-[118]. The FCS methods do not require a voltage modulation process, reducing inverter switching losses. While the CCS methods achieve vector synthesis and modulation at a fixed frequency, resulting in lower current harmonics and torque ripple.

Due to the wide application of FW control for PMSMs, this paper summarizes advanced FW control strategies, which contributes to categorizing these strategies and comparing the advantages and disadvantages from the perspectives of computational burden, control accuracy, response speed, parameter sensitivity, and control robustness. The rest of this paper is organized as follows: Section II discusses the calculation-based FW control methods. Section III analyzes the voltage closed-loop FW control strategies. Section IV introduces the MPC related methods. Section V compares the existing strategies from various aspects.

II. Calculation-based FW Control Strategies

The calculation-based FW control methods substitute the target speed and torque into the motor voltage equations and then calculate the FW current and voltage reference values. This section will first introduce the calculation for FW current trajectories, and then review the offline calculation methods based on solving algebraic equations and the online calculation methods that use mathematical methods to simplify equations and apply control algorithms to improve control accuracy.

A. Calculation of FW Current Trajectories

The d-q axis voltage equations of PMSMs are expressed as:

$$\begin{cases} u_{\mathrm{d}} = R_{\mathrm{s}} i_{\mathrm{d}} + L_{\mathrm{d}} \dot{i}_{\mathrm{d}} - \omega_{\mathrm{e}} L_{\mathrm{q}} i_{\mathrm{q}} \\ u_{\mathrm{q}} = R_{\mathrm{s}} i_{\mathrm{q}} + L_{\mathrm{q}} \dot{i}_{\mathrm{q}} + \omega_{\mathrm{e}} L_{\mathrm{d}} i_{\mathrm{d}} + \omega_{\mathrm{e}} \varphi_{\mathrm{f}} \end{cases} \quad (1)$$

where u_{d}, u_{q}, i_{d}, and i_{q} represent the d-q axis voltages and currents. L_{d} and L_{q} are the d-q axis inductances. R_{s}, ω_{e}, and φ_{f} are the stator resistance, the rotor angular velocity, and the permanent magnetic flux, respectively. The superscript "·" denotes the derivative of the variable.

The resistance and inductance voltages in (1) can usually be ignored during high-speed FW operation. The constraints of motor currents and voltages can be expressed as:

$$\begin{cases} T_{\mathrm{e}} = 1.5P\left[\varphi_{\mathrm{f}} i_{\mathrm{q}} + (L_{\mathrm{d}} - L_{\mathrm{q}}) i_{\mathrm{d}} i_{\mathrm{q}}\right] \\ i_{\mathrm{d}}^2 + i_{\mathrm{q}}^2 \le i_{\max}^2 \\ u_{\mathrm{d}}^2 + u_{\mathrm{q}}^2 = \omega_{\mathrm{e}}^2 \left[(L_{\mathrm{d}} i_{\mathrm{d}} + \varphi_{\mathrm{f}})^2 + L_{\mathrm{q}}^2 i_{\mathrm{q}}^2\right] = u_{\lim}^2 \end{cases} \quad (2)$$

where T_{e} and P represent the electromagnetic torque and the pole pairs. $u_{\lim}$ and $i_{\max}$ are the limited FW voltage and the maximum phase current, respectively.

Fig. 2 shows the control strategy division over the full speed range. As shown in Fig. 2(a), the minimum output torque of the motor is T_{e3}. When the voltage is not saturated, MTPA (OA) control is usually used, with the maximum and minimum torque points A (ω_{e1}, T_{e1}) and D (ω_{e4}, T_{e3}), respectively. After voltage saturation, the FW control is adopted, as shown in Fig. 2(b). The current trajectory moves along the current limit circle (curve AB) and the maximum torque per voltage (MTPV) boundary (curve BC), with the intersection point B (ω_{e2}, T_{e2}), and the maximum speed point C (ω_{e3}, T_{e3}). Points B and C are the tangency points of the voltage limit circle and the T_{e2} and T_{e3} torque curves. If the current moves along the AB-BC curve, the motor power remains constant, maximizing torque during speed increase. If it moves along the BE curve after point B, the motor power decreases, with the maximum speed point E (ω_{e5}, T_{e3}). In Fig. 2, the boundaries of the FW region are determined by points O, A, B, and F ($-\varphi_{\mathrm{f}}/L_{\mathrm{d}}$, 0), which represents the infinite speed point. Along the MTPA curve, the constraints of d-q axis currents are expressed as:

$$\begin{cases} \sqrt{i_{\mathrm{d}}^2 + i_{\mathrm{q}}^2} = |\boldsymbol{i}_{\mathrm{s}}| \\ T_{\mathrm{e}} = 1.5P\left[\varphi_{\mathrm{f}} i_{\mathrm{q}} + (L_{\mathrm{d}} - L_{\mathrm{q}}) i_{\mathrm{d}} i_{\mathrm{q}}\right] \end{cases} \quad (3)$$

where $\boldsymbol{i}_{\mathrm{s}}$ denotes the current vector, and β is its current angle.

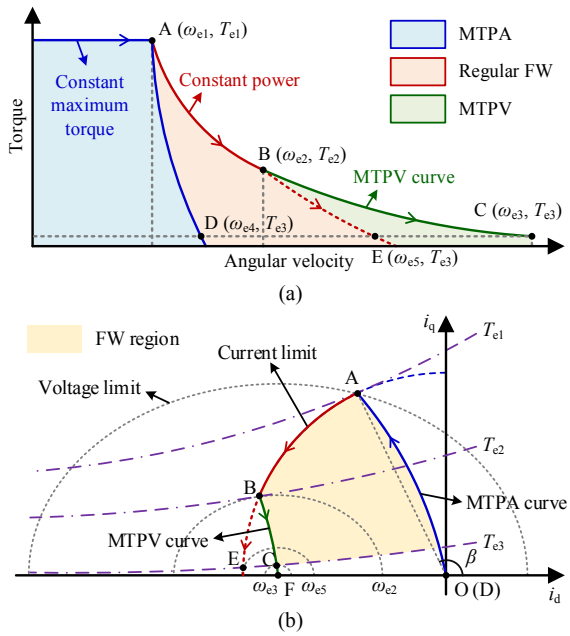


Fig. 2. Control strategies in full speed range. (a) Torque and speed perspective. (b) Voltage and current perspective.

When the current amplitude is minimum, the relationship between d-q axis currents can be expressed as:

$$i_{\mathrm{d}} = -\frac{\varphi_{\mathrm{f}}}{2(L_{\mathrm{d}} - L_{\mathrm{q}})} + \sqrt{\frac{\varphi_{\mathrm{f}}^2}{4(L_{\mathrm{d}} - L_{\mathrm{q}})^2} + i_{\mathrm{q}}^2} \quad (4)$$

Especially when the current amplitude is $i_{\max}$ at point A, β is calculated as:

$$\cos\beta = \frac{-\varphi_{\mathrm{f}} + \sqrt{\varphi_{\mathrm{f}}^2 + 8 i_{\max}^2 (L_{\mathrm{d}} - L_{\mathrm{q}})^2}}{4 i_{\max} (L_{\mathrm{d}} - L_{\mathrm{q}})} \quad (5)$$

Besides, the current constraints along the MTPV curve can be expressed as:

$$\begin{cases} \omega_{\mathrm{e}} \sqrt{(L_{\mathrm{d}} i_{\mathrm{d}} + \varphi_{\mathrm{f}})^2 + L_{\mathrm{q}}^2 i_{\mathrm{q}}^2} = u_{\lim} \\ T_{\mathrm{e}} = 1.5P\left[\varphi_{\mathrm{f}} i_{\mathrm{q}} + (L_{\mathrm{d}} - L_{\mathrm{q}}) i_{\mathrm{d}} i_{\mathrm{q}}\right] \end{cases} \quad (6)$$

When the voltage limit circle is tangent to the torque curve, the contacts form the MTPV curve. The relationship between d-q axis currents can be expressed as:

$$i_{\mathrm{d}}=-\frac{\varphi_{\mathrm{f}}}{L_{\mathrm{d}}}+\frac{-L_{\mathrm{q}}\varphi_{\mathrm{f}}+\sqrt{L_{\mathrm{q}}^2\varphi_{\mathrm{f}}^2+4L_{\mathrm{q}}^2(L_{\mathrm{d}}-L_{\mathrm{q}})^2 i_{\mathrm{q}}^2}}{2L_{\mathrm{d}}(L_{\mathrm{d}}-L_{\mathrm{q}})} \quad (7)$$

MTPA curve, current limit circle, and MTPV curve consist of the FW control boundary. With speed and torque as constants, (6) can be transformed into a fourth-order equation with d-axis current as the variable as:

$$i_{\mathrm{d}}^4+ai_{\mathrm{d}}^3+bi_{\mathrm{d}}^2+ci_{\mathrm{d}}+d=0 \quad (8)$$

where $a=\frac{2\varphi_{\mathrm{f}}}{L_{\mathrm{d}}-L_{\mathrm{q}}}(2-\frac{L_{\mathrm{q}}}{L_{\mathrm{d}}})$, $b=\frac{\varphi_{\mathrm{f}}^2}{(L_{\mathrm{d}}-L_{\mathrm{q}})^2}+\frac{4\varphi_{\mathrm{f}}^2}{L_{\mathrm{d}}(L_{\mathrm{d}}-L_{\mathrm{q}})}+\frac{\varphi_{\mathrm{f}}^2}{L_{\mathrm{d}}^2}-\frac{u_{\mathrm{lim}}^2}{\omega_{\mathrm{e}}^2L_{\mathrm{d}}^2}$, $c=\frac{2\varphi_{\mathrm{f}}}{L_{\mathrm{d}}}\left[\frac{\varphi_{\mathrm{f}}^2}{(L_{\mathrm{d}}-L_{\mathrm{q}})^2}+\frac{\varphi_{\mathrm{f}}^2}{L_{\mathrm{d}}(L_{\mathrm{d}}-L_{\mathrm{q}})}-\frac{u_{\mathrm{lim}}^2}{\omega_{\mathrm{e}}^2L_{\mathrm{d}}(L_{\mathrm{d}}-L_{\mathrm{q}})}\right]$,

$d=\frac{1}{(L_{\mathrm{d}}-L_{\mathrm{q}})^2}\left(\frac{\varphi_{\mathrm{f}}^4}{L_{\mathrm{d}}^2}+\frac{4T_{\mathrm{e}}^2L_{\mathrm{q}}^2}{9P^2L_{\mathrm{d}}^2}-\frac{u_{\mathrm{lim}}^2\varphi_{\mathrm{f}}^2}{\omega_{\mathrm{e}}^2L_{\mathrm{d}}^2}\right)$.

According to the type of the permanent magnet rotor structure, PMSMs can be divided into inner-mounted PMSMs (IPMSMs) and surface-mounted PMSMs (SPMSMs). The SPMSM can be regarded as a special IPMSM when the d-q axis inductances are equal, and the MPTA and MTPV curves in SPMSMs are simplified as i_{d}=0 and i_{d}=$-\varphi_{\mathrm{f}}/L_{\mathrm{s}}$, respectively. Whether in IPMSMs or SPMSMs, the steady-state operating points of FW control are the intersections of torque curves and voltage limit ellipses (or circles). Therefore, the FW methods applied for IPMSMs are also feasible in SPMSMs, which need to be used according to motor parameters and current limitations.

If the resistance is taken into account, the voltage limiting equation is modified as a non-standard ellipse, which can be expressed as:

$$Ai_{\mathrm{d}}^2+Bi_{\mathrm{d}}i_{\mathrm{q}}+Ci_{\mathrm{q}}^2+Di_{\mathrm{d}}+Ei_{\mathrm{q}}+F=0 \quad (9)$$

where $A=R_{\mathrm{s}}^2+\omega_{\mathrm{e}}^2L_{\mathrm{d}}^2$, $B=2R_{\mathrm{s}}\omega_{\mathrm{e}}(L_{\mathrm{d}}-L_{\mathrm{q}})$, $C=R_{\mathrm{s}}^2+\omega_{\mathrm{e}}^2L_{\mathrm{q}}^2$, $D=2L_{\mathrm{d}}\omega_{\mathrm{e}}^2\varphi_{\mathrm{f}}$, $E=2R_{\mathrm{s}}\omega_{\mathrm{e}}\varphi_{\mathrm{f}}$, $F=\omega_{\mathrm{e}}^2\varphi_{\mathrm{f}}^2-u_{\mathrm{lim}}^2$.

The center and the axis lengths of the non-standard ellipse change with the speed, rather than constant in the standard ellipse. If the non-standard ellipse is tangent to the d-axis and the current limitation is ignored, the maximum speed can be calculated as:

$$\omega_{\mathrm{emax}}=\begin{cases}+\infty & u_{\mathrm{lim}}^2L_{\mathrm{d}}^2-R_{\mathrm{s}}^2\varphi_{\mathrm{f}}^2\geq 0\\ \sqrt{\dfrac{R_{\mathrm{s}}^2u_{\mathrm{lim}}^2}{R_{\mathrm{s}}^2\varphi_{\mathrm{f}}^2-u_{\mathrm{lim}}^2L_{\mathrm{d}}^2}} & u_{\mathrm{lim}}^2L_{\mathrm{d}}^2-R_{\mathrm{s}}^2\varphi_{\mathrm{f}}^2<0\end{cases} \quad (10)$$

B. Offline Calculation Methods

The offline calculation method generates motor reference values from LUTs, which are created by solving equations, finite element analysis, and experimental verifications. The corresponding current or flux trajectories can be quickly looked up according to inputs. With accurate motor models, offline calculation methods can achieve high precision and high dynamic response FW control. Owing to low computational requirements on the MCU of electric drivers, it is widely applied in low-cost driving systems. The LUTs can be subdivided based on input variables into torque-speed tables [8]-[14], torque-flux tables [15]-[17], and multivariate tables considering motor parameters or temperature variations [18]-[21]. Since motor torque is usually difficult to obtain, it can be derived through speed proportional-integral (PI) control or torque observers.

As shown in Fig. 3(a), the d-q axis reference currents are generated by a torque-speed table through the reference torque and the feedback speed. The table is created by estimating the motor torque range and recording the current amplitudes for each torque-speed point according to the voltage equations, which is subsequently optimized by offline experiments near the steady-state operating points. In [9], the d-q axis voltages were directly calculated from the d-q axis reference currents from a torque-speed table, and the voltage vector gain was adjusted to switch to six-step operation at the maximum speed, thereby improving the voltage utilization of FW control. In [10], the influence of inverter nonlinearity was further reduced through voltage decoupling, enhancing the control accuracy of steady-state torque.

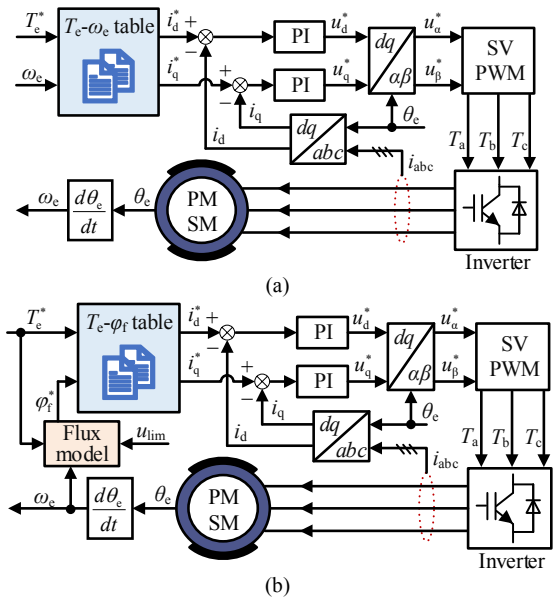


Fig. 3. Block diagram of LUT-FW control strategies. (a) Torque-speed table. (b) Torque-flux table.

Fig. 3(b) shows the block diagram of the torque-flux LUT method. Compared with the torque-speed table, it introduces the FW limit voltage and calculates the reference flux under different limit voltages, which is feasible to expand the speed and torque range of FW control. In [15], the stability under the overmodulation (OVM) strategy was improved by introducing feedback of the OVM voltage deviation to correct the reference flux. In [17], the robustness of FW control was enhanced by compensating for the observed angle deviation and correcting the d-q axis reference currents from the torque-flux table.

However, the LUTs rely on the accuracy of the motor model, especially when motor inductance, speed, torque, and position information vary during dynamic operation. Therefore, to improve the accuracy of offline calculation methods, scholars have considered motor parameter variations in the current tables and proposed multivariate LUT-FW

control strategies. In [19] and [20], the magnetic saturation on inductors was considered, and a more accurate current table with inductance changing was created. In [21], the effect of temperature on the motor was considered and a polynomial fitting method was adopted, which utilized two characteristic curves and an additional operating point to determine the optimal current operating point under multivariable conditions. LUT methods require fitting tables under various operating conditions and different parameters, presenting issues of poor generality and high labor costs. A comprehensive analysis and comparison of the three types of LUT methods are shown in Table I.

TABLE I
COMPARISON OF DIFFERENT LUT METHODS

Table input	Torque-speed	Torque-flux	Multivariate
Table feature	Simple and easy to make tables	Extension of limit voltage without making tables	High accuracy under different motor parameters
Execution	Directly LUT	Calculate flux and LUT	Calculate multivariate and LUT
Drawback	Constant voltage boundary	Indirect measure of flux vectors	Complex and hard to make tables
Reference	[8]-[14]	[15]-[17]	[18]-[21]

C. *Online Calculation Methods*

With the development of the semiconductor industry and chip manufacturing technology, small and high-performance chips are continuously being updated, significantly reducing the control costs of electric drivers. Online calculation methods solve the steady-state operating point based on motor equations, applicable to motors with different parameters and control algorithms of different structures. In field-oriented control (FOC), reference current and voltage were calculated [22]-[23], whereas in direct torque control (DTC), reference torque and flux were calculated [24]-[27]. Additionally, the block diagram of the FW method in DTC is shown in Fig. 4, where the switching states are determined by a table according to the information of the rotor position, and the substitution comparators for torque and flux errors. Compared with offline calculation methods, online calculation methods have stronger generality, avoiding the process of creating LUTs, but they also increase the system's computational burden and the control accuracy is affected by the model accuracy. Therefore, research on online calculation methods can be divided into process optimization to simplify calculation and result optimization to improve accuracy.

Due to the high-order and nonlinear characteristics of motor state equations, the solutions for current or flux vectors have multiple values and complex forms. The reference current [28]-[30] or current angle [31] needs to be calculated by dividing the FW region. Optimizing the solution process by mathematical methods can reduce computational burden and decrease the control system's usage of chip registers, as shown in Fig. 5(a), which can be divided into model simplification, polynomial fitting, and numerical iteration. In [32] and [33], the motor state equations were simplified and the solution process was converted into explicit computations that require only feedback speed and phase current information to calculate the output voltage. In [34] and [35], the d-axis current was calculated by injecting virtual voltage and using a self-learning controller, and the q-axis current was generated based on the torque equation, avoiding the solution of high-order equations. On this basis, a polynomial fitting method was used in [36] to fit the offline calculation solution into an online approximate solution for the steady state. Besides, numerical iteration methods can achieve high-precision

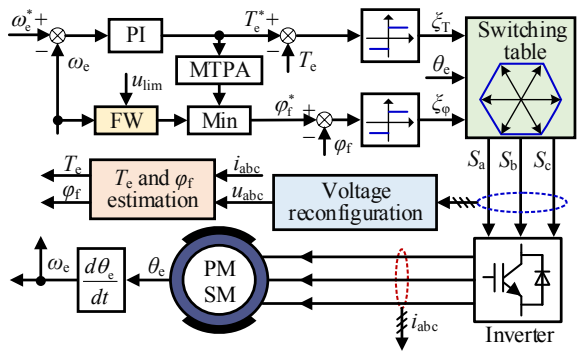


Fig. 4. Block diagram of the FW method in DTC.

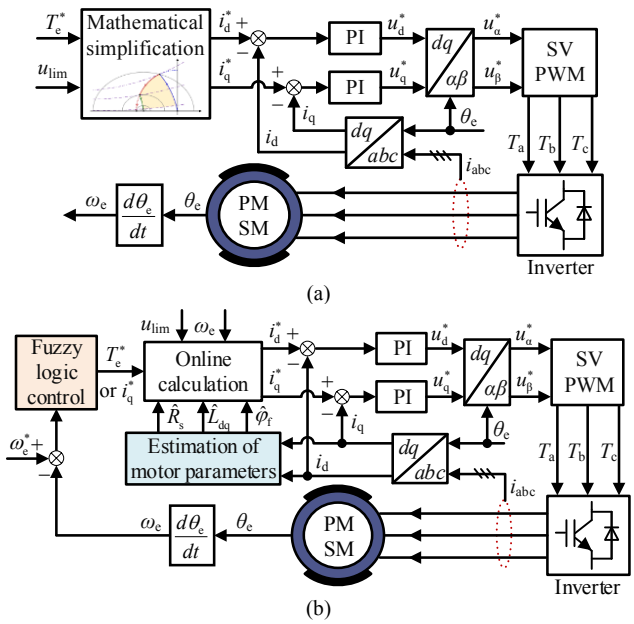


Fig. 5. Block diagram of online calculation-based FW control strategies in FOC. (a) Computational optimization. (b) Accuracy optimization.

approximate solutions after not many linear computations. In [37], the equations were linearized, and the changes in torque and voltage amplitude were calculated by current increments relative to the actual operating point, which achieves current iteration by geometric optimization. In [38], the effect of stator resistance on iteration was considered, and an iteration law on the MTPV curve was designed, achieving high-precision and wide-range FW control. To improve iteration

efficiency, the Newton iteration methods [39]-[41], the gradient descent methods [42]-[44], and the neural network iteration method based on cloud training [45] were adopted, which optimize the reference current calculation process considering computational burden, iteration number, and convergence accuracy.

On the other hand, when the motor model is inaccurate or external disturbances exist, there will be deviations between the online solutions and the ideal solutions. As shown in Fig. 5(b), parameter identification and fuzzy control methods can be used to optimize the computational results by model correction and active disturbance rejection, respectively, which improves the control accuracy and robustness. There are typically two strategies to obtain the dynamic motor parameters, which are divided into online estimation and LUT. In [46]-[49], the d-q axis inductances and flux linkage were estimated from the feedback currents or speed. The estimating method has good versatility, which can be applied to various PMSMs. However, this method usually depends on filters and needs a steady operating condition for a relatively long time, which is weak in dynamic response. To address significant variations of control boundaries caused by inductance changes, the LUT method is usually adopted to adjust the reference currents dynamically. In [50], the q-axis inductance was fitted by the q-axis current to modify the reference currents. Considering the coupling effect, the d-q axis inductances were both fitted to two-dimensional planes [51]. Based on this, the changing voltage drops on inductors and the constant voltage drops on resistors were tabulated in [52] and [39], respectively, which improved the dynamic response and the accuracy of the reference currents. Therefore, the control boundaries can still be divided accurately if the motor parameters are updated in time by online estimation or LUT. In [53] and [54], fuzzy control was used to obtain the d-q axis reference currents, enhancing the robustness of FW control when the motor model mismatches. Compared with open-loop process optimization methods, result optimization methods often introduce feedback control, thereby achieving stronger control robustness, and the comparisons of these methods are shown in Table II.

TABLE II
COMPARISON OF DIFFERENT ONLINE CALCULATION METHODS

Type	Computational optimization			Accuracy optimization	
Method	Model simplification	Polynomial fit	Numerical iteration	Parameter estimation	Fuzzy control
Feature	Exact solution by less feedback information	Approximate solution by offline data	Approximate solution by initial value	Improve model accuracy	Improve control robustness
Execution	Directly calculate voltages by reference values	Calculate FOC reference currents by polynomial fit	Calculate FOC reference currents by numerical iteration	Online estimate flux, inductance, and resistance	Adaptively compensate for model mismatch
Drawback	Complex numerical computation	Depend on offline solutions of FW currents	Depend on convergence analysis and iteration cycles	Depend on offline parameter table or signal injection	Depend on fuzzy rules
Reference	[32]-[35]	[36]	[37]-[45]	[46]-[51]	[53], [54]

III. VOLTAGE CLOSED-LOOP FW CONTROL STRATEGIES

Voltage closed-loop FW control strategies adjust the FW current by comparing the output voltage with the limit voltage, which does not depend on motor parameters. These methods are simple in structure and strong in control robustness, which can be divided into dual-regulator control and single-regulator control according to the control dimension.

A. Dual Regulator Methods

The dual-regulator control is consistent with MTPA control, allowing for stable and smooth switching between the two control strategies [56]-[57]. The most widely used method is the d-axis current regulation method [58]-[60]. As shown in Fig. 6(a), the d-axis current, which weakens the permanent magnetic flux, is obtained by a PI controller. Research on dual-regulator FW control methods can be summarized into three main directions: application of OVM strategies, controller parameter design, and disturbance suppression.

To improve the utilization of the DC-bus voltage, FW control strategies are usually combined with OVM strategies, expanding the torque and speed range by increasing the limit voltage amplitude or duty cycle [61]. However, this inevitably

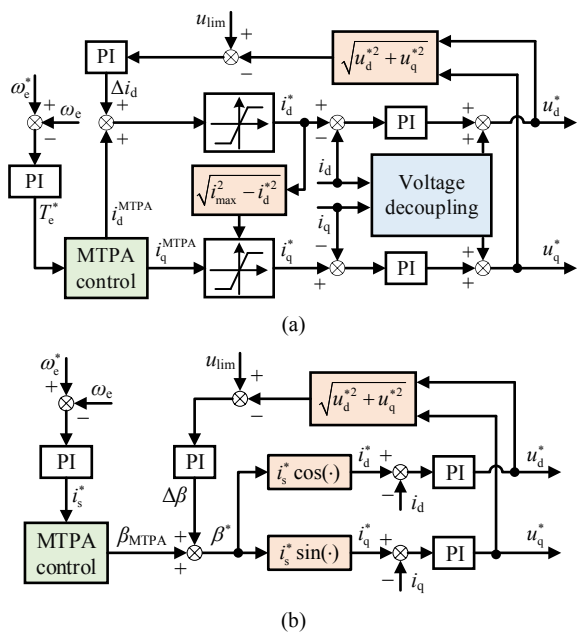


Fig. 6. Block diagram of voltage closed-loop FW control strategy by dual regulators. (a) D-axis current regulation. (b) Current angle regulation.

leads to increased current harmonics and output voltage saturation [62]. To balance the voltage utilization and current

harmonics of the OVM strategy, a PI controller was added in [63] and [64] to adaptively adjust the OVM voltage boundary. In [65], the OVM region was divided into four quadrants based on motor load and speed requirements, where the output voltages were designed correspondingly. In [66], the FW limit voltage was optimized based on a power model of speed and torque. To address the issue of output voltage saturation, three methods are generally summarized. The first method involves correcting the FW input voltage or duty cycle. The difference between the limited and pre-limited d-axis voltage in [67], the difference between the limited and pre-limited d-q axis duty cycles in [68], and the difference between the OVM and pre-OVM q-axis voltage in [69] were introduced into the FW current regulation respectively, preventing the saturation of PI controller in the current loop. The second method involves correcting the FW output current. In [70] and [71], the difference between the limited and pre-limited FW current from the FW PI controller was fed back, reducing the FW current overshoot. The third method involves correcting the regulation method of the FW current. In [72], the difference between the OVM and pre-OVM q-axis voltage was calculated, and a low-pass filter and a proportional link were used to generate the FW current, which achieves anti-saturation FW control. In [73], the d-axis voltage saturation was also considered, the difference between the OVM and pre-OVM d-q axis voltages were both introduced into the FW current regulation. On this basis, the high-frequency components of the q-axis voltage difference were compensated into the d-axis current in [74], improving the dynamic performance of the FW current.

The optimization of controller coefficients is usually from time domain or frequency domain perspectives, focusing on motor control performance and system transfer function characteristics, respectively. From the time domain perspective, the integral coefficient was designed based on the rise time and changing rate of FW current in [72], [75]-[76], improving the response speed of FW current and reducing overshoot. In [68] and [71], the particle swarm and genetic algorithms were used, respectively, which offline optimize PI coefficients to reduce the error of anti-saturation FW controllers within the OVM range. The dual-regulator adjusts the current reference values through voltage closed-loop control, with its output voltage fluctuating around the FW limit voltage, inevitably leading to coupling issues between the inner current loop and the outer voltage loop. In [64], the FW control coefficients were comprehensively adjusted based on the experimental performance of voltage ripple, current response time, and overshoot, which improves the dynamic performance and avoids voltage saturation. From the frequency domain perspective, the control coefficients were set according to the bandwidth of the FW transfer functions in [69]-[70], [77], achieving a balance between dynamic response and steady-state disturbance rejection. In [78], the transfer function of the voltage closed-loop FW control system was analyzed based on a small-signal model, and the stable FW control was achieved by configuring the poles of the characteristic equation.

When the motor runs at high speed, non-ideal factors such as inverter nonlinearity, DC-bus voltage, and load fluctuations can also degrade the performance of FW control. In [73], voltage deviations caused by stator resistance and inverter nonlinearity were considered, and the q-axis current was compensated by distorted voltage reconstruction, improving the dynamic performance of FW control. In [79], feedback voltage fluctuations caused by small voltage margins in the FW region were analyzed, and the speed controller was optimized using fuzzy control, reducing steady-state speed and torque ripples. In [80], the effect of DC-bus voltage ripples on FW control stability was analyzed in electrolytic capacitorless drive systems, and current ripples were reduced through harmonic torque compensation. In addition, the quasi-steady-state control, feedforward control based on reference dynamics, and error feedback control were employed in [59] to calculate torque under nonlinear pressure loads, improving the response speed and control accuracy. For sensorless control systems with periodic load disturbances, a switching mode adaptive angle error compensator was proposed in [60], reducing speed ripples in FW control. To address the issue of non-ideal back EMF, a space vector decoupling stage was added to the voltage closed-loop FW control in [81], which suppresses the 5th and 7th harmonics in the FW current. In [82], the speed overshoot during current trajectory transitions was analyzed and a variable virtual current constraint method was proposed to achieve both current and speed steady-states, reducing motor speed adjustment time and speed overshoot.

To move the current trajectory along the maximum current circle, current angle adjustment is another FW control approach. As shown in Fig. 6(b), the voltage closed-loop control with dual regulators remains for smooth switching and high control robustness, and the current angle is adjusted through PI control to correct the current trajectory [83]-[85]. In [86], the dynamic performance in current angle control was improved during six-step operation by introducing adaptive controller gain. In [83] and [86], the transfer function from the current angle to output voltages was established based on a small-signal model, and the controller coefficients were optimized according to the stability margin of the transfer function's amplitude-frequency characteristics, which ensures the stability within the whole FW range. In summary, voltage closed-loop FW control strategies can be divided into two main categories: d-axis current regulation and current angle regulation, which are compared in Table III.

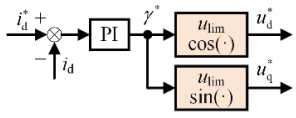
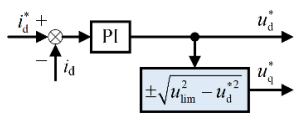
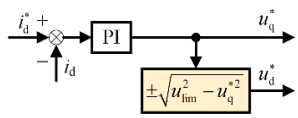
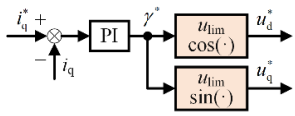
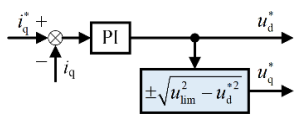
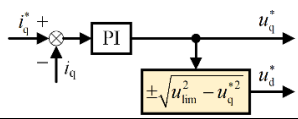
B. Single Regulator Methods

In the FW region, if resistance and inductance voltage drops are ignored, the motor's steady-state operating point is the intersection of the voltage limit circle and the reference torque curve, and the control system's freedom degree can be reduced to 1 [87]-[89]. According to the combination of FW control inputs (d-axis or q-axis current) and outputs (voltage angle, d-axis or q-axis voltage), six basic single-regulator FW control methods are summarized in Table IV. These methods take advantage of the constant amplitude of the output voltage in the FW region, and directly decouple the d-q axis voltages,

TABLE III
COMPARISON OF DIFFERENT DUAL-REGULATOR BASED METHODS

Term / Type	OVM execution	Control coefficients design	Disturbance suppression	Input variable
D-axis current regulation	1. Voltage boundary adaptation [63]-[66] 2. Voltage anti-saturation [67]-[74]	1. Time domain [64], [68], [71], [72], [75]-[76] 2. Frequency domain [69]-[70], [77]-[78]	1. Inverter nonlinearity [73] 2. Voltage ripple [79]-[80] 3. Nonlinear load [59], [60] 4. Back EMF harmonic [81] 5. Speed overshoot [82]	1. Voltage [56], [58]-[60], [62]-[67], [69], [72]-[82] 2. Speed [57] 3. Duty cycle [61], [68], [70]-[71]
Current angle regulation	Adaptive voltage gain [86]	Frequency domain [83], [86]	Not mentioned	Voltage [83]-[86]

TABLE IV
BLOCK DIAGRAM OF DIFFERENT SINGLE-REGULATOR BASED METHODS

Input / Output	Voltage angle	D-axis voltage	Q-axis voltage
D-axis current			
Q-axis current			

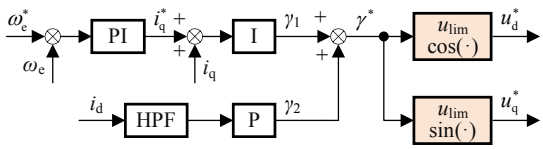


Fig. 7. Block diagram of voltage angle regulation FW control strategy in [90].

rather than use anti-windup methods in the dual-regulator based methods, which contributes to addressing the issues of the coupling of the inner current loop and the outer voltage loop and the saturation of the current regulators in a simple control structure.

Research on single-regulator FW control mainly focuses on improving control system stability and suppressing current or torque ripples. As shown in Fig. 7 of [90], the base voltage angle was calculated through q-axis current integration, and the dynamic angle was compensated by d-axis current after high-frequency filtering, which improves the control stability. In [91], the effect of d-axis current feedback on the transfer function poles of the voltage angle regulation FW control system was analyzed, which verifies the effectiveness in suppressing torque ripples. Moreover, the current harmonics were analyzed from the system impedance in [92], and an offline parameter optimization method was proposed based on amplitude-frequency characteristics and step response, which suppresses current harmonic under motor parameter mismatch conditions. The effect of motor parameter mismatch on voltage angle regulation was further analyzed in [93], and a dual-parameter compensation method was proposed to improve the dynamic performance of the speed loop. In [94], a feedforward compensation was applied to the voltage angle by speed and voltage information, improving the stability of voltage angle regulation. In [95], the limitations of the FW region in voltage angle regulation FW control for unlimited-speed motors were analyzed, and a single-regulator FW control using d-axis current to regulate q-axis voltage was proposed to expand the FW control boundaries. Based on these, six single-regulator FW control structures were reviewed and compared in [96] according to the pole distribution and theoretical stability range, verifying the actual stability range of different control structures through experiments.

IV. MPC RELATED METHODS

To solve the issue of complex nonlinear constraints within each control cycle, MPC related methods have been widely investigated [97]. The core idea of these methods is to construct a cost function based on predictions and constraints and calculate the output variables that minimize the cost function. The cost function for FW control typically consists of speed, torque, current, and flux. Based on the characteristics of the output voltage vectors, these methods can be divided into FCS and CCS.

As shown in Fig. 8(a), FCS-MPC does not include a modulation stage, allowing for a lower switching frequency, which reduces switching losses. In [100], model predictive direct torque control and FW control were combined, and the optimal voltage vector was selected from a finite voltage set through the cost function, achieving high-speed operation at a low switching frequency. In [101], a second-order Kalman filter was used to estimate load disturbances in model predictive speed control, achieving good dynamic speed performance under inaccurate motor mechanical parameters. Besides, a cost function in model predictive flux control was constructed in [102], and the inverter voltage and current limitations were applied in the high-speed region on the flux plane, which directly reflects the saturation of motor flux. To

address the FW current trajectory deviation issue caused by an inaccurate motor voltage model, a virtual voltage FW scheme was designed in [103] to correct the FW current trajectory based on motor model characteristics, predicted current errors, and disturbance current information.

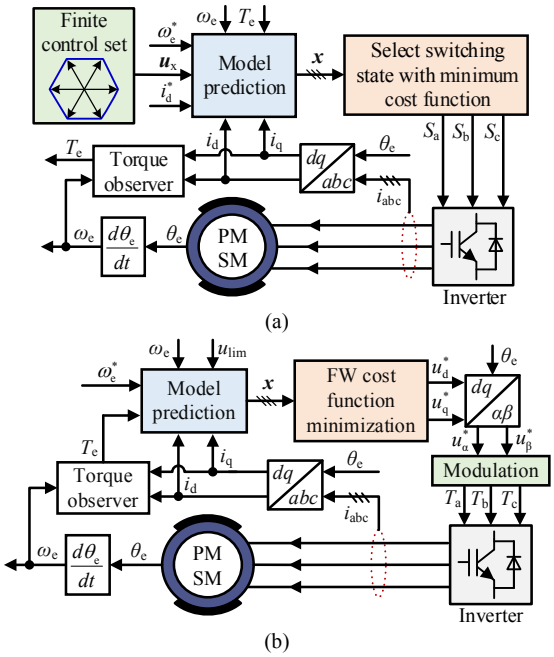


Fig. 8. Block diagram of MPC related methods. (a) FCS. (b) CCS.

In Fig. 8(b), CCS-MPC adds a voltage modulation stage with a fixed carrier frequency, resulting in smaller stator current harmonics [104]. In [105], d-q axis currents, speed, or rotor position were used as variables for explicit calculation, where a first-order error integrator and a second-order error integrator were adopted to construct the cost function, respectively. In [106], an explicit calculation method based on a linearized model was proposed, and the output voltage was compensated to eliminate the error between the predicted and actual speeds. Based on this, the linear model was optimized in [107] considering viscosity coefficients and motor saliency, and the amplitude of the linear output voltage was constrained to reduce output torque overshoot. Besides, the performance control algorithm was adopted in [108] to correct the model speed, enhancing the dynamic accuracy of model prediction. In [109], the effect of different sampling times of speed and current loops on all-time prediction results was analyzed, and a linear estimation method based on virtual time was proposed, improving the performance of fast prediction times for slow sampling models. In addition to torque, speed, and current, flux can also be used to achieve precise predictive results. In [110], a cost function using stator flux was constructed, and the dual flux vectors were optimized to change the flux angle within each control cycle. To reduce the calculation dependence on permanent magnetic flux for the stator flux vector, d-q axis flux linkages were considered in [111] to construct the cost function, and the flux disturbances were compensated using a Luenberger observer.

In [112], a direct speed MPC with state-dependent Riccati equations was proposed, which transformed FW current and voltage into a quadratic programming problem and allowed online calculation of the optimal output voltage based on the given speed. To quickly and efficiently solve the quadratic programming problem, a dual active set solver was used in [113], and two standard matrix update methods were combined to reduce the burden on the MCU. In addition, deep neural networks were applied in [114], and disturbance compensation was adopted to perform offline learning for the neural networks based on the standard motor model, which improves the performance of long-horizon MPC. Since the voltage limit in model prediction is usually set to a constant value, current harmonics increase when the motor operates in the OVM region. To address this issue, harmonic reference generators were used in [115]-[117], which added harmonic reference values to the average reference values to suppress current harmonics. Besides, the OVM voltage boundary in [118] was extended by constructing irregular hexagonal current and voltage constraints. Generally, the comparisons of FCS and CCS methods are summarized in Table V.

TABLE V
COMPARISON OF FCS AND CCS METHODS

Control set	FCS [100]-[103]	CCS [104]-[118]
Feature	Without modulation	With modulation
Drawback	Large torque ripple	Large switching loss
Cost function minimization	Switching state selection [100]-[103]	1. Explicit calculation [105]-[111] 2. Quadratic programming [112], [113] 3. Deep neural network [114]

V. ANALYSIS AND COMPARISON

When PMSMs operate at high speed, the increased back EMF generated by the permanent magnetic flux needs to be counteracted by FW control. Due to high performance and robustness, the online calculation methods and voltage closed-loop control methods have been more widely studied. Online calculation methods calculate reference currents and voltages based on motor models with better dynamic performance, but they also depend on model accuracy. Therefore, improving model accuracy through fuzzy control and motor parameter identification methods is an effective way to enhance the control performance. However, this inevitably increases the difficulty of solving high-order equations. Therefore, model simplification, polynomial fitting, and numerical iteration methods have also been applied to reduce the computational burden on the controller.

Offline calculation methods are more cost-effective versions of online calculation methods. The reference values

of LUT methods are created by experiments or offline calculations, which are used during motor operation. Since the LUTs are predetermined, these methods cannot adjust in real time under different operating conditions, making the motor performance more sensitive to external disturbance and motor parameters. However, in some low-cost controllers, the LUT method is the most economical FW control method. Expanding simple two-dimensional tables to multivariate tables that include motor parameter variations can effectively improve the control performance without excessively increasing the control burden.

Compared with calculation-based methods, voltage closed-loop control methods do not depend on motor model accuracy, which primarily adjusts FW current through voltage or duty cycle feedback control. The control performance depends on the FW current generator and current regulators, leading to research on optimizing controller coefficients. Furthermore, OVM strategies have broader applications in voltage closed-loop FW control strategies, and the modulation index can be directly set by the FW limit voltage. Since these strategies do not depend on the motor model, even if the reference voltage exceeds the voltage hexagon in the OVM region, it can be corrected through the voltage feedback. Therefore, voltage closed-loop control strategies can offer stronger control robustness with the anti-saturation and disturbance suppression algorithms.

Compared with the mentioned three FW control methods, MPC is an intelligent control method that has emerged into FW control recently. It relies on model iterations to predict motor state variables and calculates output variables that minimize the cost function, achieving rapid adjustment of speed and torque. From FCS-MPC, which selects inverter switch states, to CCS-MPC, which outputs duty cycles from the voltage modulation stage, the motor's torque harmonics are effectively suppressed, but the switching losses would increase. Moreover, expanding the cost function's output variables from speed and torque to current and flux can achieve higher precision and faster response control, but it also significantly increases the computational burden for minimizing the cost function. Theoretically, the speed range of the MPC related methods is similar to that of other FW control methods. However, the issues of motor model mismatch and prediction delay would result in prediction error and limit the speed range, which need to be compensated by specific strategies. Besides, the MPC related methods avoid the integrated saturation for the coupling system regulated by PI controllers and have the advantage of handling nonlinear constraints, which contributes to improving the control dynamics. However, the current ripple suppression methods based on frequency domain analysis cannot be directly applied to MPC methods because of the nonlinear control law. The suppression strategy based on a harmonic reference generator inevitably increases the computational burden Therefore, balancing control performance and computational cost has always been important research in MPC.

The comparisons of the four FW control strategies are summarized in Fig. 9. From the perspective of calculation burden, the online calculation and MPC methods are heavier because of solving high-order equations and multiple model predictions. From the perspective of control robustness, the voltage closed-loop method is satisfactory because these methods utilize feedback control and do not depend on motor parameters. In contrast, the online calculation methods are based on motor equations and are more sensitive to motor parameters. From the perspective of response speed, the offline calculation method is best because the reference variables are obtained by LUT without any calculation, which is poor in generalization and robustness. From the perspective of control accuracy, the MPC methods have a greater advantage because the controller can handle complex nonlinearities and constraints by cost functions.

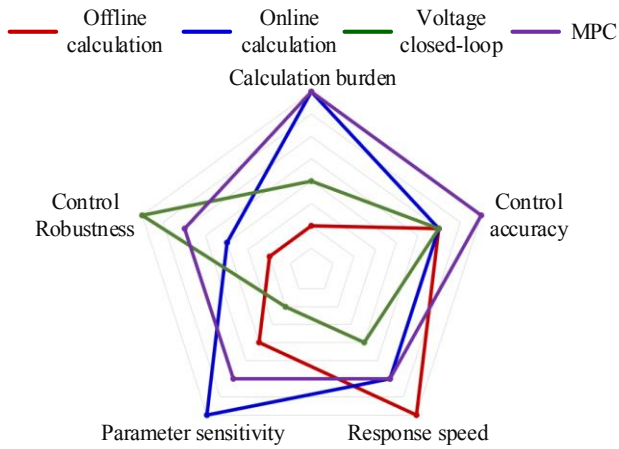


Fig. 9. Comparisons of four FW control strategies.

VI. CONCLUSION

Currently, FW control strategies for PMSMs have been extensively researched and applied, providing theoretical foundations and implementation methods for high-speed motor drive control. Considering factors such as controller cost, operating conditions, inverter non-ideal characteristics, and motor model uncertainty, there is still a need for further research into the high performance and robustness of FW control. This paper analyzes recent FW control strategies for PMSMs, categorizing the above strategies into three main types from the perspective of control theories, and summarizing the advantages and disadvantages of each type of FW strategy. Finally, these FW control strategies are compared in terms of computational and control performance. In summary, offline calculation methods, online computational methods, and MPC all depend on model accuracy but offer good current response speed. Online calculation methods and MPC are mainly applied on platforms with strong computational capabilities, while offline calculation methods are mainly used in low-cost industrial products. Voltage closed-loop control methods do not depend on motor parameters and have relatively simple control structures, achieving a balance between computational difficulty and control robustness. Future research directions for FW control mainly focus on simplifying the computational process of high precision models and improving the dynamic

performance of high robustness closed-loop control algorithms.

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